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DEVELOPMENT OF 3D MODEL SLICER FOR A 5-AXIS 3D PRINTER

1. Background and Objective

Additive manufacturing using conventional three-axis 3D printers can encounter difficulties when fabricating models with significant overhanging geometries. Such features commonly require additional support structures, increasing material consumption and post-processing requirements.

Five-axis 3D printing offers an alternative by allowing the printing orientation to change during fabrication, potentially enabling overhanging structures to be printed with reduced support requirements. However, the five-axis 3D printer developed for this project did not have dedicated slicer software capable of generating toolpaths for its specific configuration.

This research aimed to develop a dedicated 3D model slicer for the five-axis printer, enabling model processing, orientation planning, and toolpath generation for five-axis printing.

2. Methods

The research followed an iterative Research and Development (R&D) approach. The work began with studying conventional slicing methods and investigating algorithms that could potentially be adapted for five-axis printing.

Different algorithmic approaches were analyzed and tested using STL models to identify their capabilities and limitations. Based on these investigations, a segment-based slicing approach was developed and progressively improved through software testing, G-code verification, and physical printing experiments.

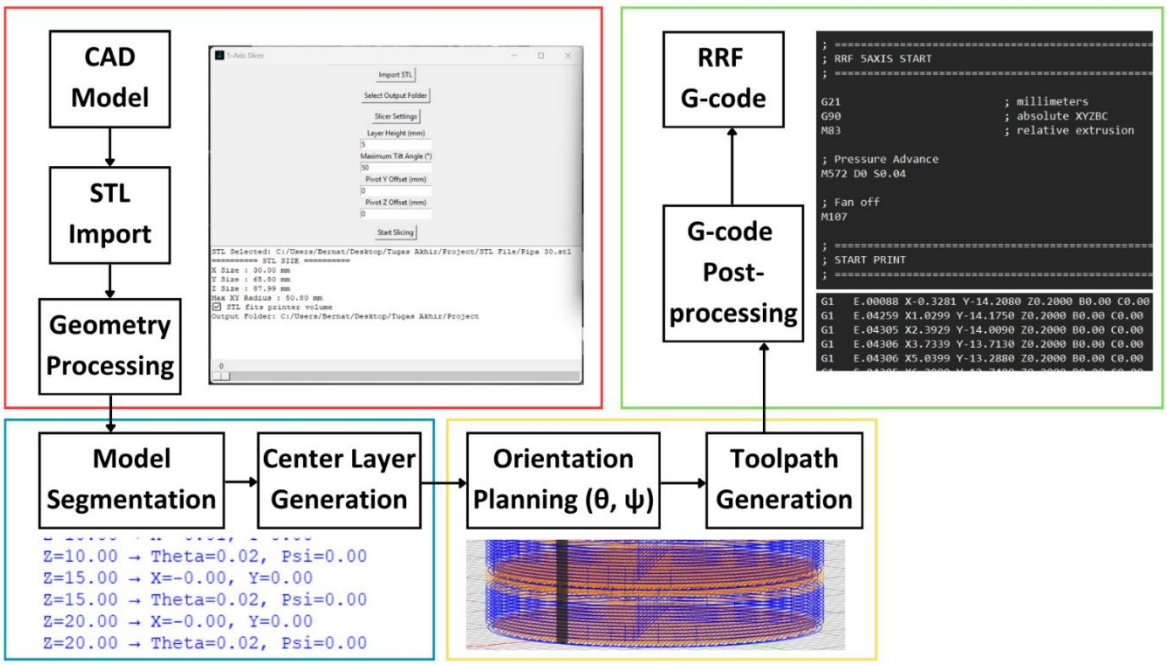


Figure 1. Developed Five-Axis Slicer Software Workflow

The developed software consists of the following main processing stages, as illustrated in Figure 1:

1. Model Import and Geometry Processing: Importing CAD-derived STL models and processing their geometry.
2. Model Segmentation: Dividing the model into segments according to changes in orientation.
3. Center Layer Generation: Determining representative center positions for the generated layers.
4. Orientation Planning: Calculating the required printing orientation using the layer center positions.
5. Toolpath Generation: Generating printing paths based on the processed geometry and planned orientations.
6. G-code Post-processing: Transforming the generated G-code into machine-specific coordinates and preparing the output for the five-axis printer.

The software was independently developed by the researcher, including the algorithm investigation, slicing implementation, G-code transformation workflow, and graphical user interface. Physical testing was conducted in collaboration with a teammate responsible for the printer hardware.

3. Results and Discussion

The developed slicer successfully generated non-planar toolpaths for five-axis printing and demonstrated its capability to fabricate overhanging geometries without additional support structures in the tested cases. One example is the successful printing of a bent solid cylinder using the five-axis printer, as shown in Figure 2.



Figure 2. Support-free fabrication of a solid cylinder with a 30° bend using the developed five-axis slicer.

The dimensional deviation between the original STL design and the physical specimen was 0.95% for the five-axis printer, compared with 0.885% for a Bambu Lab A1 Mini print produced with support structures. These results provide a reference for comparing the dimensional accuracy of the tested specimens under the experimental conditions.

Unlike the rotation-driven deformation and quaternion-field approach used in S^3-Slicer, this research employed a segment-based orientation strategy to generate non-planar slicing paths from STL models.

However, the developed approach has an important limitation. Orientation planning relies on representative center positions of individual Z-layers. Consequently, a layer is represented by a single dominant orientation. For geometries containing multiple branches with different orientations, this representation cannot adequately capture the independent orientations of each branch.

This limitation restricts the applicability of the current algorithm to more complex branching geometries and indicates the need for further improvements in geometry analysis and orientation planning.

4. Research Contribution

The primary contribution of this research was the independent development of a dedicated slicer software for a five-axis 3D printer. The work covered algorithm investigation, development of the segment-based slicing method, software implementation, G-code transformation, and graphical user interface development.

The project demonstrated the feasibility of generating non-planar toolpaths for five-axis additive manufacturing while identifying limitations in the representation and processing of complex multi-branch geometries.

Full Thesis: The complete undergraduate thesis, including detailed methodology, implementation, and experimental results, is available in my portfolio: https://ber-sento.github.io/